Critical Thinking

From the NACE competencies, critical thinking was the one used the most throughout my capstone project. With no prior experience in electronic components, I had to do extensive research in voltage and current limits, as well as determining how to connect components safely to ensure the entire system would function correctly. This included 45 kg-cm servos, motor drivers, and the Arduino system. This took careful thinking of the system as a whole, from the power distribution from the 12V 3S battery through the UBEC voltage converters into each component, accounting for the maximum accumulated current that can occur when differing systems are activated. This needed to be documented to ensure that the system could not overheat or short. Since this was the first time I have tackled such a large project, it was really beneficial in teaching me how to take a large, foreign problem and break it down piece by piece to put together a solid solution. Critical thinking has been one of my most relied-on skills, and this capstone has been an amazing opportunity to allow for this competency to grow. It will be crucial for my success throughout my engineering career.

Technology

This project significantly helped me advance my technological competency. I have never previously used a power distribution block, nor directly wired or soldered any system together. Besides just wiring, I also got to expand my coding and signaling directory knowhow to fit an Arduino brain up to my project, ensuring I would be able to completely control every subsystem as desired. This took a ton of research through Arduino forums to find the best applications for each of my components. The best example of this being applied comes from learning how to bridge the gap from hardware to software. These skills are transferable to embedded systems, robotics, and any career involving hardware to software integration.

Communication

My capstone was anything but a solitary task to apply new and old technical skills. It required a lot of comprehensive communication between my groupmates and sponsor. Original contact for this project was through Dr. Oman and the MIME program, as they acted as initial contact and financial sponsor for the project. After many design brainstorming meetings with my group, we were able to identify Curtiss-Wright as a potential manufacturing sponsor through a groupmate's personal connections. After an entire debrief with Curtiss-Wright of both the project and design parameters, it was determined that they would be able to help us and use their professional equipment to manufacture components for our bot. Our group continued this communication to make design alterations to multiple components to ensure our design met offensive and defensive design parameters, as well as manufacturability. Because of this, we were able to get a waterjet aluminum claw, a bent sheet metal ramp, custom CNC servo horns, and a custom carbon-printed servo moment arm.

Professionalism

Taking on the sole responsibility for the entire electrical system of my capstone's bot demanded a lot of professionalism. This took a great deal of attention to detail and accountability. I got to really see the relevance of this when a small mistake that I made during testing resulted in damaged components and a delay in testing. I took full responsibility for this mistake, personally reordered the damaged parts, and made all the repairs right away to correct this. Being in charge of an important system in a project, and following through to ensure that the team's success was able to be maintained, have been important skills I have gained from completing my capstone project. This accountability and professional reliability are skills I continue to utilize moving forward with not only my career, but in life as well.

Career & Self-Development

The SUMOBOT capstone was deliberately chosen by me to challenge and expand all my existing skills. There are many things, including wiring diagrams, electrical components, carbon printing, and waterjet manufacturing, that have not crossed my plate in my engineering education. Rather than staying within my comfort zone of CAD, I was able to apply my current skillset to a whole new set of design problems, creating solutions I wouldn't have had the chance to previously formulate. In my career I wish to pursue devices that have a little bit of everything when it comes to mechanical engineering, and I was really lacking on the code and electronic side of this. Jumping into a project that combines it all and forcing myself to create all of it from scratch has been a priceless experience and has pushed me to learn so much more. I now can tackle any sort of electronic component and use technology to communicate efficiently with these components to complete any desired task.